

TELESTO AI PLATFORM ARCHITECTURE

Technical Specification & Workflow Engineering Report • TBOP Core Foundation

1. Executive Overview & Problem Statement

Small and Medium Enterprises (SMEs) in emerging retail markets currently operate under a **fragmented manual architecture** comprising social messaging apps (Facebook Messenger, WhatsApp), unstructured voice calls, manual spreadsheets, and physical notebooks. This fragmented operational state creates critical operational bottlenecks: missing customer inquiries during peak hours (slow replies), manual inventory checks leading to stockouts or discrepancies, and zero systematic tracking of customer purchase history.

Operational Diagnosis & Platform Remedy

Current State Bottlenecks: FB Messenger + WhatsApp + Phone + Spreadsheets + Notebooks

- *Slow Customer Replies:* Inquiries missed during peak sales hours.
- *Stock Mismanagement:* Manual paper tracking leads to frequent stockouts.
- *Zero Customer History:* No systematic repeat follow-ups or retention tracking.

Telesto AI Solution: Unify sales, inventory, and customer chats into a single AI platform, shifting SME operations from manual delays to an automated 24/7 intelligent engine.

2. Multi-Tier Platform Architecture

The Telesto AI Platform is architected as a modular, enterprise-grade system built on the **Telesto / TBOP Core Enterprise Engine**. The multi-tiered topology abstracts system complexity behind a simplified SME interface while orchestrating unified data, workflow engine execution, and multi-modal AI capabilities.

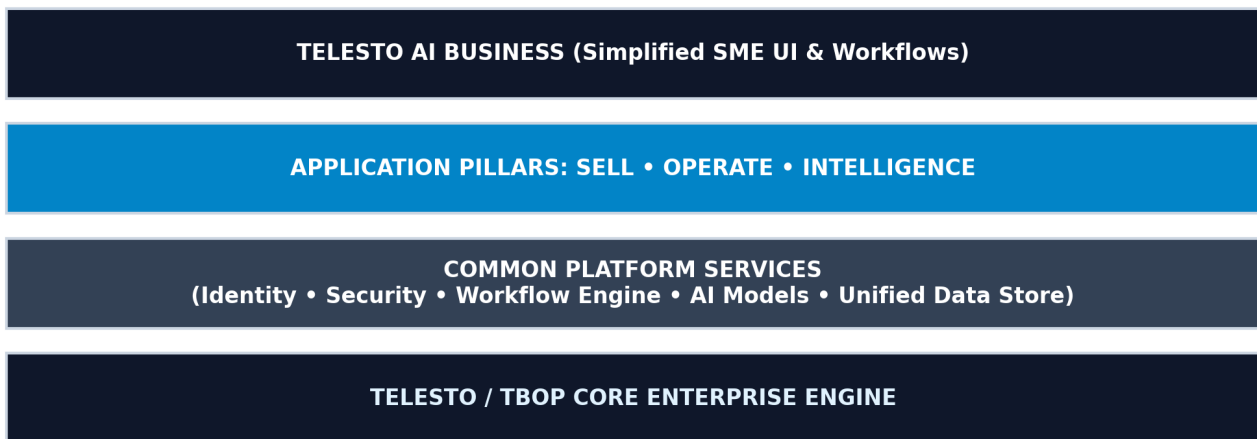


Figure 1: Layered Platform Architecture Stack (TBOP Enterprise Foundation)

Layer	Component Services	Architectural Function
User Interface	TELESTO AI BUSINESS	Simplified SME dashboard and unified messaging workspace.
Application Pillars	SELL • OPERATE • INTELLIGENCE	Core commercial engines for growth, operations, and business insights.
Shared Platform	Common Platform Services	Identity, Security, Workflow Engine, AI Models, Unified Data Store.
Foundation Engine	TBOP Core Enterprise	Enterprise-grade core infrastructure enabling high-concurrency scalability.

3. Functional Pillars & Capability Matrix

Telesto operationalizes SME growth through three interconnected pillars: **SELL** (Growth & Marketing), **OPERATE** (Streamlined Operations), and **INTELLIGENCE** (AI Business Copilot).

Pillar	Core Capability	Operational Features & Data Touchpoints
SELL (Growth & Marketing)	Revenue Acquisition & Engagement	- AI social marketing content generation - Unified customer CRM database - Automated social campaign generator & lead tracking 24/7 AI chat responder across Messenger & WhatsApp
OPERATE (Streamlined Operations)	Order Execution & Stock Lifecycle	- Instant digital order creation from chat - Real-time size/variant inventory tracking (SKU matrix) - Payment status logging (COD/bKash/Nagad) - Automated delivery status & stock deduction
INTELLIGENCE (AI Business Copilot)	Executive Guidance & Analytics	- Executive performance dashboard - Automated low-stock alerts & threshold triggers - Demand prediction models & sales forecasting - Natural language conversational business analytics

4. Boutique Workflow Transformation Specification

The workflow transformation automates the end-to-end commercial path from customer social inquiry to fulfilled digital order. Figure 2 illustrates the automated 5-step transaction sequence.

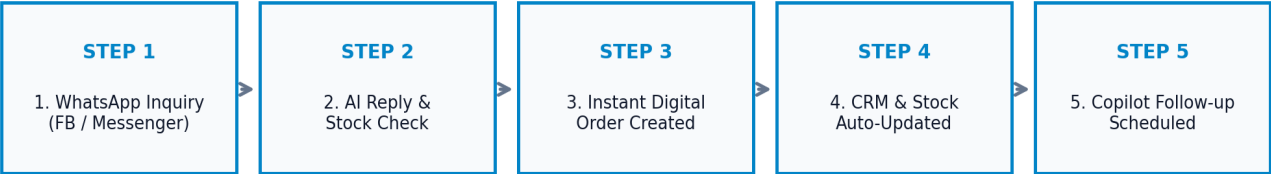


Figure 2: 5-Step Automated Transaction Sequence Flow

Step	Trigger & State	Automated Logic & API Execution	System Result
Step 1	Customer Social Inquiry	Customer sends message via FB Messenger/WhatsApp requesting price/size.	Ingested by AI Message Listener.

Step	Trigger & State	Automated Logic & API Execution	System Result
Step 2	Instant AI Reply & Stock Check	AI parses inquiry intent, queries Variant Inventory for stock, and crafts response.	Instant 24/7 quote sent to customer.
Step 3	Instant Order Creation	Customer confirms buy; AI generates Digital Order with item, size, and COD terms.	Digital Order record issued.
Step 4	Automated CRM & Stock Update	Atomic inventory decrement executed; customer profile logged into Unified CRM.	Stock reduced, CRM profile created.
Step 5	Copilot Follow-Up & Alerts	AI Copilot schedules automated post-delivery check-in & checks stock thresholds.	Retention trigger set, alert logged.

5. Data Schemas & Unified Data Store

The Unified Data Store enforces transactional consistency across inventory variants, customer CRM, and order states using atomic database constraints.

Entity	Primary Keys & Fields	Data Types & Constraints	Relational Mapping
crm_customers	id, phone, name, channel, orders_count, total_spent_bdt	VARCHAR(50), UNIQUE(phone), INTEGER, DECIMAL(10,2)	1-to-Many with orders
variant_inventory	variant_id, product_name, category, size, color, stock_qty, price_bdt, safety_threshold	VARCHAR(50) PK, VARCHAR(100), INTEGER, DECIMAL(10,2)	Referenced by orders
orders	id, customer_id, variant_id, quantity, total_price_bdt, payment_status, delivery_status	VARCHAR(50) PK, FK(crm_customers), FK(variant_inventory)	Many-to-1 Customer & Variant
copilot_schedules	id, customer_id, order_id, scheduled_date, status	VARCHAR(50) PK, FK(orders), DATE, VARCHAR(20)	1-to-1 Order Follow-up

6. Targeted Sector Configurations

The core Telesto platform provides specialized configuration profiles optimized for high-priority Bangladeshi SME commercial sectors.

Sector Profile	Primary Commercial Focus	Configured Feature Modules
Boutiques & Fashion	Apparel, Jamdani, Panjabi, Festive Wear	- Size & color variant matrix tracking - Automated festive social media marketing campaigns
Handicrafts & Jute (e.g., Julhas Handicrafts)	Artisanal Crafts, Jute Goods, Export Goods	- Storytelling marketing generator - Digital export product catalogues
Jewelry & Gifts	High-Value Personal Accessories, Custom Gifts	- High-value customer CRM profile tracking - Automated repeat customer loyalty tracking
Retail & Food	Fast-Moving Consumer Goods, Bakery, Groceries	- Rapid order creation workflow - Real-time automated low-stock safety triggers

Platform Architectural Synthesis

Architecture Conclusion: By pairing the TBOP Core Enterprise Engine with modular Common Platform Services and SME-tailored Application Pillars (SELL, OPERATE, INTELLIGENCE), Telesto delivers a scalable enterprise architecture that transforms Bangladeshi SME retail operations from delayed manual workflows into a highly efficient, automated digital ecosystem.